Metric completion need not preserve minimal Sobolev gradients

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Abstract

A doubling metric measure space supporting a Poincaré inequality has a function with source minimal gradient at most 1 and completion gradient $\sqrt{2}$. The completion complement is nonmeasurable. The conclusion concerns exactly the catalogue target; no historical priority claim is made.

Catalogue scope. AIM 669; complete resolution. Canonical target at revision `1e56463`. The corrected completion-measure convention is also discussed in [1, Section 4].

1 Statement and argument

The statement in the catalog allows the original metric space to be nonmeasurable in its completion. Under exactly this allowance the answer is negative. The construction below uses the axiom of choice, but does not use the continuum hypothesis. The measure hypotheses are those of [2]: a positive complete Borel measure, finite and nonzero on balls. Compact inner regularity is not an additional hypothesis there.

Theorem 1.1. *For every $1 < p < \infty$ there is a globally doubling metric measure space X supporting a global 1-Poincaré inequality, whose metric and measure completion is the Euclidean plane with Lebesgue measure, and a bounded Lipschitz function u on a ball $\Omega \subset X$ such that*

$$g_{u,\Omega} \le 1 \quad \mu\text{-almost everywhere}, \qquad g_{\widehat{u},\Omega^\wedge} = \sqrt{2} \quad \mu\text{-almost everywhere on } \Omega$$

for every Dirichlet-space extension $\widehat{u}$ agreeing with u quasieverywhere. Thus the constant $A_0 = 1$ in Problem 669 is impossible.

1.1 A set excluding oblique curves

Lemma 1.2. *There is a set $A \subset \mathbb{R}$, containing 0, such that both A and $\mathbb{R} \setminus A$ meet every positive-measure Borel subset of $\mathbb{R}$, and every rectifiable curve in*

$$X = (A \times \mathbb{R}) \cup (\mathbb{R} \times A)$$

has an axis-parallel tangent at almost every arclength parameter.

Proof. Call a Lipschitz map $\gamma : [0,1] \to \mathbb{R}^2$ oblique if

$$\mathcal{L}^1\{t : \gamma_1'(t) \neq 0 \text{ and } \gamma_2'(t) \neq 0\} > 0.$$

*The solutions and partial results in this document were found by AI. The supplied research and the separate repository review are AI work; human peer review and formal verification are not claimed.

There are at most $\mathfrak{c}$ such maps, since continuous maps are determined by their values on a countable dense set. There are exactly $\mathfrak{c}$ positive-measure Borel subsets of $\mathbb{R}$.

For each oblique map there is a positive-measure set $S \subset [0,1]$ on which both coordinate functions are injective. Indeed, choose $\delta > 0$ so that the set of differentiability points satisfying $|\gamma_i'(t)| \geq \delta$, $i = 1, 2$, has positive measure. At each such point, for all sufficiently small nonzero h with $t + h \in [0,1]$,

$$|\gamma_i(t+h) - \gamma_i(t)| \geq \tfrac{\delta}{2}|h|, \qquad i = 1, 2.$$

The measurable subsets on which this holds for every $0 < |h| < 1/m$, $m \in \mathbb{N}$, cover that set. One of them has positive measure. Intersect it with an interval of length less than $1/m$ having positive-measure intersection. The resulting set S satisfies the displayed inequality for every pair of its points. In particular each coordinate is injective on S, and $|S| = \mathfrak{c}$.

Well-order, in one list of cardinality $\mathfrak{c}$, all the oblique maps and all positive-measure Borel subsets of $\mathbb{R}$. Maintain disjoint sets of real numbers assigned to A and to B, initially assigning 0 to A. At a Borel-set task choose two previously unassigned points of that set, assigning one to A and one to B. At an oblique-map task choose $t \in S$ such that both $\gamma_1(t)$ and $\gamma_2(t)$ are previously unassigned, and assign both to B.

These choices are always possible. At a stage indexed by an ordinal $\alpha < \mathfrak{c}$, fewer than $\mathfrak{c}$ numbers have been assigned. Every positive-measure Borel set has cardinality $\mathfrak{c}$. At a curve task, coordinate injectivity on S implies that fewer than $\mathfrak{c}$ parameters are prohibited by earlier assignments. This argument does not require $\mathfrak{c}$ to be a regular cardinal.

After all tasks assign every remaining number to B, and let A be the set assigned to A. The Borel-set tasks give the first assertion. Every oblique map has a selected point in $B \times B$, so cannot have its whole image in X. Apply this observation to arclength parametrizations of rectifiable curves, affinely rescaled to $[0,1]$, to obtain the second assertion. □

1.2 The trace measure

Equip X with the Euclidean subspace metric. The set A is dense, so $\widehat{X} = \mathbb{R}^2$. Every positive-measure ambient Borel set meets $A \times \mathbb{R}$, hence meets X: by Fubini, the set of first coordinates with positive-measure vertical section has positive measure and therefore meets A. Thus $\mathbb{R}^2 \setminus X$ has inner Lebesgue measure zero.

Define a measure on the trace Borel sigma algebra by

$$\mu(E \cap X) = \mathcal{L}^2(E), \qquad E \subset \mathbb{R}^2 \text{ Borel}, \tag{1}$$

and complete it. This is well-defined: if two ambient Borel sets have the same trace, their Borel symmetric difference is contained in $\mathbb{R}^2 \setminus X$ and is null. If traces are pairwise disjoint, their representatives have null pairwise intersections; disjointifying the representatives proves countable additivity. Every ball satisfies

$$\mu(B_X(c,r)) = \pi r^2, \qquad \mu(B_X(c,2r)) = 4\mu(B_X(c,r)).$$

The completion measure prescribed in the question is consequently ordinary completed Lebesgue measure.

We will use two elementary properties of this trace construction. Every μ-null set is contained in $N \cap X$ for some ambient null Borel set N. Also, every μ-measurable real-valued function has an ambient Lebesgue-measurable representative agreeing with it outside such a trace null set. To see the latter, first take a Borel version for the completed trace measure. A real-valued Borel function on a subspace has an ambient Borel extension: construct it from Borel extensions of simple-function approximations and their pointwise limits. Trace integrals then equal ambient integrals by (1) and simple-function approximation. These are also the measure-extension conventions in the corrigendum to [2].

1.3 A global Poincaré inequality

We prove the 1-Poincaré inequality directly; Hölder's inequality then gives the p-Poincaré inequality for every $p \geq 1$. Let $B = B_X(c, r)$, let v be integrable on $2B$, and let g be a Borel upper gradient of v there. We may suppose $g \in L^1(2B)$, since otherwise the required inequality is trivial. Choose ambient Borel representatives V, G on the Euclidean ball $\widehat{B}(c, 2r)$, with $G \geq 0$, and one ambient null Borel set N outside whose trace both representatives agree with the original functions. Let $Q = I_1 \times I_2$ be the open axis-aligned square of side $2r$ centered at c. Then

$$\widehat{B}(c, r) \subset Q \subset \widehat{B}(c, 2r).$$

By Fubini, outside a null set $Z \subset I_2$ the functions $V(\cdot, y)$ and $G(\cdot, y)$ are integrable on I_1 and the section N_y is null. For every $y \in (A \cap I_2) \setminus Z$, the whole horizontal chord $I_1 \times \{y\}$ lies in X. Thus, for every pair $s, t \in I_1 \setminus N_y$, the upper-gradient inequality gives

$$|V(s, y) - V(t, y)| \leq \int_{\min(s,t)}^{\max(s,t)} G(x, y)\, dx. \tag{2}$$

This gives an absolutely continuous representative of the slice and $|\partial_1 V(\cdot, y)| \leq G(\cdot, y)$ almost everywhere. For example, absolute continuity of the integral on the right first extends V from the full-measure set $I_1 \setminus N_y$ continuously to I_1; applying (2) to finitely many disjoint intervals proves absolute continuity and the derivative bound.

The passage from the possibly nonmeasurable family A to almost every line is justified as follows. Fix a countable family $\mathcal{T} \subset C_c^\infty(I_1)$ dense in the C^1 topology on each of a countable compact exhaustion of I_1. The set of parameters $y \in I_2 \setminus Z$ for which at least one inequality

$$\left| \int_{I_1} V(x, y) \varphi'(x)\, dx \right| \leq \int_{I_1} G(x, y) |\varphi(x)|\, dx, \qquad \varphi \in \mathcal{T}, \tag{3}$$

fails is Lebesgue measurable, by Fubini and countability. By the previous paragraph it is disjoint from A. It must therefore be null: every positive-measure measurable set contains a positive-measure Borel set, and A meets all such sets. Density extends (3) to every smooth compactly supported test function. The one-dimensional distributional characterization of absolute continuity gives $|\partial_1 V| \leq G$ on almost every horizontal slice. Repeating the argument on the vertical lines whose first coordinate is in A gives $|\partial_2 V| \leq G$ on almost every vertical slice.

Fubini and integration by parts on the slices now show that $V \in W^{1,1}(Q)$ and

$$|\nabla V| \leq \sqrt{2}\, G \quad \text{almost everywhere on } Q.$$

The usual Euclidean L^1 Poincaré inequality on $\widehat{B}(c, r)$, followed by the ball-volume ratio, therefore gives

$$\fint_B |v - v_B|\, d\mu \leq Cr \fint_{\widehat{B}(c,r)} |\nabla V|\, d\mathcal{L}^2 \leq C' r \fint_{2B} g\, d\mu.$$

The constants are absolute and independent of B, v, g. This proves the claimed global Poincaré inequality with dilation factor 2.

1.4 The gradient gap

Take $B_0 = B_X(0, 2)$ and $\Omega = B_X(0, 1)$. The definition $\Omega^\wedge = \widehat{X} \setminus \overline{X \setminus \Omega}$ gives the ordinary open Euclidean unit disk, since A is dense. Set $u(x_1, x_2) = x_1 + x_2$ on Ω. Every rectifiable curve in Ω, parametrized by arclength, has at most one nonzero coordinate derivative almost everywhere. Consequently

$$|(u \circ \gamma)'| = |\gamma_1' + \gamma_2'| = |\gamma'| \quad \text{almost everywhere.}$$

Thus $g = 1$ is an upper gradient of u, belongs to $L^p(\Omega)$, and $g_{u,\Omega} \leq 1$ almost everywhere.

Suppose $\widehat{u} \in D^p(\Omega^\wedge)$ agrees with u outside a set of zero C_p^X capacity. Zero Sobolev capacity implies zero μ-measure, since the capacity includes the L^p term. The disagreement set is therefore contained in $N \cap X$ for an ambient null Borel set N. Since X meets every positive-measure ambient Borel set, the measurable function $\widehat{u} - (x_1 + x_2)$ must vanish Lebesgue-almost everywhere on $\Omega^\wedge$. Indeed, a positive-measure set of disagreement outside N would contain a positive-measure Borel set meeting X, a contradiction.

Euclidean Sobolev theory identifies the minimal p-weak upper gradient with the length of the weak gradient. It follows that

$$g_{\widehat{u},\Omega^\wedge} = |\nabla(x_1 + x_2)| = \sqrt{2} \quad \text{Lebesgue-almost everywhere.}$$

By the trace-measure identity this also holds μ-almost everywhere on Ω. This proves the theorem. In this example any valid comparison constant A_0 is at least $\sqrt{2}$.

The construction addresses the exact generality of the catalog. It does not assert failure when X is required to be measurable in its completion.

References

[1] A. Björn, J. Björn and P. Lahti, *Removable sets for Newtonian Sobolev spaces and a characterization of p-path almost open sets*, Rev. Mat. Iberoam. **39** (2023), 1143–1180. https://arxiv.org/abs/2105.09012v2. DOI: https://doi.org/10.4171/RMI/1419.

[2] A. Björn and J. Björn, *Poincaré inequalities and Newtonian Sobolev functions on noncomplete metric spaces*, J. Differential Equations **266** (2019), 44–69; corrigendum, **285** (2021), 493–495. https://arxiv.org/abs/1705.02253; https://doi.org/10.1016/j.jde.2021.03.008. DOI: https://doi.org/10.1016/j.jde.2018.07.029.